

Deadlines You Can Docket Against

Verifying Computed Trademark Deadlines Against 300,000 Register Records

Signa Research

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Abstract

Most trademark data errors cost analysis quality. Deadline errors cost the trademark. A missed renewal or maintenance filing does not degrade a registration; it cancels it. The calendars that prevent this (docketing systems) are traditionally maintained by hand, office by office, at considerable expense. Signa computes these deadlines: renewal cycles, declarations of use, grace periods, restoration windows, and opposition periods, from statutory rules exposed directly in the API. At the v1.0 evaluation those rules covered 22 jurisdictions; the rule set has grown since (see `GET /v1/deadline-rules`).

This report does two things. First, it describes the engine and the standard we hold it to: deadline computation at Signa is a pure function (no database, no network, no clock dependence beyond an explicit as-of date), every rule carries citations to its statutory sources with a last-verified date, and the full rule corpus is itself published through the API for inspection. Second, and mainly, it reports a register-based verification. We froze a pre-registered, hash-manifested sample of 31,547 registrations across 10 production offices, then an expanded same-seed sample (stability check) of 306,896 that is a strict superset of the first (the same seed, so every original record is retained), and measured our computed deadlines against three complementary forms of ground truth: dates the offices themselves publish on the record (for one office, USPTO, the compared date is connector-derived; that cell is flagged and its independent evidence is the event arm), the dated renewal and maintenance filings that actually occurred in register histories, and 34,132 observed opposition proceedings. The rules were corrected against these same records during the evaluation, so the expanded sample is a stability check, not an independent holdout.

The engine does not guess. Where a record does not carry the input the statute requires (three documented cohorts, section 3.4) and the office has stated no expiry to anchor on, it declares that the office's date is required instead of computing a date that the register showed to be wrong a material share of the time. This policy is specified in the product code and applied identically in the evaluation, so nothing below depends on an after-the-fact choice of population.

On the committed 31,547-record sample, every comparable record is either verified or declared: 99.68% (95% CI 99.60 to 99.73) of 29,295 comparable records are either computed and exactly equal to the office-published date, or explicitly declared as requiring the office's date. On the records the engine computes (26,800, or 91.5% of the comparable population), exact agreement between the pure statutory computation, with the office's date withheld, and the office-published date is 99.65% (95% CI 99.57 to 99.71); weighted by each stratum's true production share it is

99.82%. The schedule the API actually serves, which anchors on the office's stated date where one exists, reproduces that date for 29,292 of 29,295 records (99.99%); because that path uses the office's date as an input it is a description of what customers receive, not independent evidence of rule correctness. 94.7% of 24,091 observed renewal and maintenance filings on computed records fall inside our computed windows (99.7% at USPTO), 92.7% of all 27,562 on the served schedule, and 91.0% of 23,751 verifiable opposition filings fall inside our computed opposition windows (99.7% at EUIPO, the cleanest cell). Every disagreement was individually classified by mechanism; 95 records disagree after the declared cohorts are set aside, and the genuinely unexplained residual remains 3 records. The v0.91 full-population figure of 96.08% at expanded scale, which scored the engine's guesses on the declared cohorts against it, is retained in section 5.1 for continuity.

The verification also found defects: six in our own rules (five in the maintenance rules, one in the opposition rules), all fixed, with agreement reported before and after each fix (a seventh reported in earlier versions, on the Swedish 2019 transition, is withdrawn: section 5.4); three in our data ingestion (two fixed, one filed); and 311 records classified by Signa as register-data errors, with per-record evidence, where the register rather than the computation carries the wrong date. A subsequent mutation-testing hardening pass over the rules package (version 1.0, section 3.6) found and fixed a further ten engine defects, none of which moved a verified date on the ten offices. We publish the methodology in full, including the discrepancy taxonomy and the cases we got wrong, because a deadline product asks for a specific kind of trust, and we believe that trust should be checkable.

About this report. This is a first-party evaluation: Signa selected the metrics, wrote the evaluator, corrected the system under test, and classified the disagreements. It has not been independently audited. The artifacts are published so that any reader can check the work.

1. Background: the docketing problem

Registered trademarks are not granted once; they are kept alive. Nearly every jurisdiction requires periodic renewal, and several require affirmative proof that the mark is actually in use: the United States demands a declaration of use between the fifth and sixth year after registration (Section 8 of the Lanham Act), again at each renewal (Section 8 combined with Section 9), and under a separate provision for Madrid-route registrations (Section 71). Miss the window, and after a short grace period the registration is cancelled. Oppositions run on even tighter calendars: a third party typically has a fixed window after publication to oppose, and a brand owner watching for conflicts has exactly that window to act.

What makes this hard to automate is not any single rule but the accumulation of jurisdictional texture:

- **Different triggers.** Renewal terms run from filing in some jurisdictions and from registration in others. Sweden switched triggers on 2019-01-01, and the correct trigger for a Swedish mark depends on whether its application was made before that date.
- **Route matters.** The same US registration is governed by Section 8 or Section 71 depending on whether it arrived directly or as a Madrid designation, and a Madrid

designation's renewal anywhere runs on the international registration's ten-year cycle, not the national trigger.

- **Legacy regimes.** Canadian registrations renewed before the 2019-06-17 reform carry 15-year terms and can sit on mixed 15-and-10-year ladders. Australian marks accepted before 2013-04-15 had a three-month opposition window; later ones have two.
- **Calendar arithmetic is legally defined.** What "three months after" means at a month boundary is specified by statute and differs by country (German law, the Canadian Interpretation Act, the Singapore Interpretation Act, and EU delegated regulation all define it, not identically). Leap days, month-end clamping, end-of-month rules for older German registrations, and weekend or holiday rollover (37 CFR 2.196 in the US, Singapore's excluded-day practice including the eves of three holidays) all change real dates.
- **A deadline is a legal day, not a timestamp.** An opposition window closes at the end of the office's legal day in the office's time zone. A system that treats it as a UTC timestamp will tell a California user that a European deadline closed hours before it legally did.

Law firms and corporate IP departments solve this with commercial docketing systems and manual verification. Developers building trademark features have had no API-native option: general trademark data APIs return raw dates from the register and leave the statutory arithmetic to the caller.

1.1 Why verification, not just testing

The engine described here is extensively unit-tested against hand-derived golden dates (section 3.6). But hand-derived goldens share a failure mode with the rules they test: both are produced by humans reading statutes. The strongest available check is the register itself: offices publish expiry and renewal-due dates on records, register histories contain the dated renewal and maintenance filings that actually happened, and opposition proceedings carry the dates oppositions were actually filed. If our computed deadline for a real registration disagrees with what the office printed on that registration, one of us is wrong, and either answer to that question is worth publishing. That comparison, at scale, across offices, is this report.

2. What is computed

The engine produces two families of output.

Maintenance deadlines. For a given mark (jurisdiction, filing route, status, and the record's date fields), the engine returns the schedule of lifecycle deadlines: renewal (recurring by statutory cycle, including legacy-term ladders where a reform changed the cycle length), declaration of use (US Sections 8 and 71), combined renewal and use filings (US Section 8 with Section 9), declaration of incontestability (US Section 15, the one optional entry),

restoration windows following a missed deadline, and international-registration renewal for Madrid records. Each computed deadline carries the date the window opens, the due date, the grace-period expiry where one exists, a status relative to an explicit as-of date (future, window open, due soon, in grace, missed), an urgency tier, and the consequence if missed (cancellation, expiration, or lapse of the international registration).

Opposition windows. For a published application or registration, the engine computes the statutory opposition window: trigger event (publication for opposition, registration publication, or international-designation publication, depending on the office), open and close dates, the routinely available extension where office practice has one, and a must-act-by moment expressed as the end of the office's legal day in the office's time zone with daylight saving handled. Each computed window carries the citation and version of the rule that produced it.

3. The engine

3.1 Rules as audited data, not buried code

Rules are data. At the v1.0 evaluation there were 23 jurisdiction configurations covering 22 jurisdictions for maintenance deadlines (the United States carries two, one for direct registrations and one for Madrid designations), and 27 opposition-window rules covering both direct and Madrid-designation routes at the covered offices. The rule set has grown since: in October 2026 it holds 30 maintenance configurations for 29 jurisdictions and 48 opposition-window rules (see `GET /v1/deadline-rules`). Every configuration carries citations to its statutory and office sources, each with the date it was last checked, and a last-verified date for the configuration as a whole. Computed opposition windows embed the rule version that produced them, so any alert or API response can be traced to the exact rule text it came from.

The rule corpus is not a secret. It is published, with citations, through the API (`GET /v1/deadline-rules` and `GET /v1/opposition-rules`), so a customer or their counsel can audit the rules we apply before relying on them. The full deadline rule corpus is published for inspection.

3.2 Purity as a correctness strategy

Deadline computation is a pure function: no database access, no network calls, and no hidden dependence on the current clock (callers pass an explicit as-of date; statuses and urgencies are computed relative to it). Identical inputs always produce identical output. This is a correctness strategy, not an implementation detail: pure functions can be property-tested exhaustively, reproduced exactly in an investigation, and recomputed identically at read time, so there is no stored deadline to go stale when a rule is corrected.

Dates are treated as calendar dates, not timestamps, except at the one place time zones legally matter: the must-act-by moment of an opposition window, which is computed as the end of the office's legal day in the office's own time zone. Date arithmetic follows the applicable legal definitions: month addition clamps to month-end where statutes say so, leap days resolve the way the relevant interpretation acts specify, older German registrations get the end-of-month rule, pre-reform Canadian registrations get the legacy-term ladder, the Swedish trigger change applies to applications made from its statutory date, and deadlines roll forward over weekends and office holidays against the office holiday calendars (24 at the v1.0 evaluation) maintained for 2024 through 2030 (including, for Singapore, the office's excluded-day eves).

3.3 The office's own dates as an anchor

Two behaviors connect the statutory computation to what the office has already said about a record. First, when the office states an expiry date and the statutory ladder computed from the record's trigger date passes through it exactly, the office has confirmed the computation and the full trigger-derived schedule is used. When the stated expiry does not sit on the computable ladder (because a discriminating date is absent from the record, as with Madrid subsequent designations, or because a legacy cadence predates the modeled regime), the schedule is re-anchored on the office's stated date. Second, a renewal cycle that fell due before the office-stated expiry has by definition already been paid (that is why the term now runs to the stated expiry), so it is not shown as an actionable deadline. Both behaviors are measured separately in section 5: the pure statutory computation is evaluated with the office's dates withheld, precisely so that the evaluation cannot be circular.

3.4 Declared computability (version 1.0)

Some records do not carry the input the statute needs, and no amount of arithmetic recovers it. Three cohorts are documented: Australian direct marks filed before 1996-01-01, where the 1955 Act's seven-and-fourteen-year cadence versus the 1995 Act's ten-year term turns on a registration date the register records as equal to the filing date, so roughly one in six of the cohort is on the modern ladder and indistinguishable from the rest; Singapore direct marks filed before 1999-01-15, on the repealed Cap. 332 cadence, with the same problem; and Madrid designations whose row carries no international-registration date, where the renewal term runs on the IR's anniversary (Madrid Protocol Art. 6(1) and 7(1)). For the third cohort the engine first looks up the WIPO parent registration we ingest (by IR number) and anchors on its date; this is exact for original and subsequent designations alike. Where no parent is available, a designation's own filing or registration date is used only at offices where this evaluation verified that it equals the IR date (near-100% agreement per office: cipo, nipo, ipau, ipi and wipo on the filing date; euipo and prv on the registration date). Elsewhere, notably IPOS, whose priority-linked international marks carry neither column as the anchor, no stand-in is used.

When the office states an expiry date, the anchored schedule of section 3.3 is exact and the record is computed. Otherwise the engine declines: it returns no schedule and reports `requires_office_date` with the cohort, on `POST /v1/deadlines/compute` as supported: `false` with that reason. This replaced the previous behavior of computing a guess for these records, and it replaced the previous evaluation's post-hoc "modeled population": the same rule now lives in the product and in the harness, so the population the engine is scored on is the population it claims to compute. A declared record is scored as correct behavior (no wrong date was served) and reported separately; it is never counted as agreement.

3.5 Exposure and cost

Computed deadlines appear on the trademark detail endpoint (`GET /v1/trademarks/{id}`), on portfolio deadline views (including an iCalendar feed), and inside watch alerts, where the opposition must-act-by is computed with the provenance fields attached. For bulk and integration use, two dedicated endpoints compute without persisting anything: `POST /v1/deadlines/compute` and `POST /v1/oppositions/compute` accept up to 1,000 items per call, take an explicit as-of date and horizon, and return per-item results including an explicit supported: `false` with a reason for jurisdictions we have not modeled, rather than a silent guess. The engine computes full schedules for the entire 31,547-record evaluation sample in roughly 256 milliseconds single-threaded on a laptop, about 8 microseconds per record.

3.6 Standing tests and the frozen regression gate

The rules package ships with roughly 1,450 test cases: 31 plain-language specification files (one per jurisdiction plus the engine mechanics, each scenario an executable test), per-office golden dates for maintenance and opposition computations, golden pins over every statutory configuration, calendar-arithmetic and holiday-rollover suites, and property-based tests executed across every jurisdiction configuration. In August 2026 the package was put through mutation testing: 2,946 mutants across all 57 files, every one killed by a test or classified in place with a reason (100%); the same was done for the compute endpoint. That pass found ten engine defects, all fixed and re-verified against the frozen sample with no movement on the ten verified offices; they are listed in the changelog.

A frozen regression gate runs in continuous integration on every change: it recomputes the 31,547-record maintenance sample and pins that sample's Arm A agreement and declared counts, its Arm B totals, and every Arm C cell of the opposition sample at the frozen as-of date, so an engine change that moves a single record fails the build until it is re-verified cell by cell. The gate does not cover the expanded 306,896-record run or the production-weighted outputs.

4. Verification methodology

4.1 Design principles

Four commitments govern the design. The sample was frozen (record identifier lists hashed with SHA-256) before any comparison ran. Every number carries its n , headline numbers carry 95% confidence intervals (Wilson), and no cell below $n=50$ is reported. Every disagreement is individually classified by mechanism, not left in an "other" bucket. And when the evaluation itself exposed a defect in a rule, we corrected the rule and report agreement both before and after the correction. No correction was accepted on empirical fit alone; each required a statutory or documented-data-source justification, verified independently against primary sources, and was required to fit the register data without regressing any other cell. Several implementation choices (anchor selection, era boundaries) were informed by the data and are disclosed as such. Correcting rules against the evaluation and then reporting only the corrected number would make the exercise circular; showing both makes a correction evidence of the process working. One protocol amendment is disclosed rather than left silent: the pre-registered comparison horizon of 15 years was raised to 45 years during the evaluation after it produced false disagreements on old Canadian records (section 5.7); a wider horizon cannot create false matches.

4.2 Samples

Three frozen datasets are used.

- **Maintenance sample.** A deterministic, seeded sample of 31,547 registrations across 10 production offices (USPTO, EUIPO, WIPO, CIPO, IP Australia, IPOS, the Norwegian, Swedish, and Swiss offices, and INPI France), stratified by filing route (direct versus Madrid designation) and registration era at 500 records per stratum, to exercise legacy-regime rules deliberately.
- **Expanded same-seed sample (stability check).** The same design at tenfold depth: 306,896 records, drawn with the same seed, yielding 282,393 comparable records. Because the seed is identical, this is a strict same-seed superset of the development sample (every original record is retained), not an independent holdout: the rules were corrected against these same records. It checks whether the headline rates are numerically stable at greater depth, not whether the corrected system generalizes to unseen records.
- **Opposition sample.** 34,132 opposition proceedings with observed filing dates across the five offices whose proceedings data carries oppositions (USPTO, EUIPO, CIPO, IP Australia, IPOS), hash-ordered per office and route cell.

Records with statuses under which no deadline is defined were excluded by pre-registered rule. Because equal-quota strata overweight disagreement-heavy populations (legacy eras, Madrid designations at specific offices) relative to production, the headline is also reported

reweighted by each stratum's true production population (9,166,622 records). The weighted Arm A figures and the pooled Arm B and Arm C figures do not yet carry design-based uncertainty intervals; a simple binomial interval would understate the uncertainty because events and comparisons cluster within records and strata, so those intervals are queued as future work.

4.3 Three arms of ground truth

- **Arm A, office-published dates.** For offices that state an expiry or renewal-due date on the record, the computed next renewal deadline must equal the office's stated date exactly. Disagreements are additionally banded by distance. Weekend and holiday filing rollover is treated as a filing-day convention, not an expiry disagreement. The office's stated expiry is withheld from the engine input in this arm, so the computation being tested cannot see the answer (section 3.3). Under the declared- computability policy (section 3.4) a record the engine declines is reported as declared, counted as correct behavior in the verified-or-declared figure, and excluded from the agreement denominator; it is never counted as agreement.
- **Arm B, event consistency.** For offices whose register histories carry dated renewal and maintenance filings, every observed filing must fall inside the window we computed for it (window open through grace expiry). USPTO events are restricted to a pre-registered allowlist of filing-coded prosecution events, excluding courtesy reminders and office notices, which measure office latency rather than rule correctness. Events on declared records are not scored in the pure arm (the engine claims no window for them) and are counted; the served-schedule figure scores all events.
- **Arm C, observed oppositions.** Every observed opposition must have been filed inside the computed opposition window for the opposed record's publication date, counting the rule's modeled routine extension separately. One non-circular validity filter applies: proceedings data attaches an opposition to both parties' records, and a record already registered when the opposition was filed cannot be the opposed application, so those rows are excluded by registration timing alone.

Two circularity guards matter for reading Arm A. USPTO publishes no expiry date (Signa derives one), so only actually-renewed USPTO records enter Arm A and the USPTO cell is the weakest in the table by construction; the USPTO evidence that matters is Arm B. And where an office publishes a renewal-due date it equals the expiry date in every feed except WIPO, so it is not an independent signal and Arm A uses the stated expiry only.

5. Results

All numbers are reproducible from the committed frozen data and results artifacts; runs are identified by date in the changelog of the evaluation record.

5.1 Headline

All figures below are from the committed 31,547-record sample re-run on 2026-10-04 (version 1.1) under the version 1.0 policy (section 3.4), except where marked vo.91.

Metric	Value	95% CI	n	Notes
Verified or declared (exact agreement + declared, over all comparable records)	99.68%	[99.60, 99.73]	29,295	primary figure
Pure statutory computation, office's date withheld, computed records	99.65%	[99.57, 99.71]	26,800	engine correctness on the population it claims
Pure statutory computation, production-weighted	99.82%		estimated over 8,965,731 production records	
Records declared (requires the office's date)	2,495		8.52% of 29,295	AU direct pre-1996: 363; IPOS direct pre-1999: 592; IPOS Madrid designations: 1,540
Schedule the API serves, equal-quota	99.99%	[99.97, 100.00]	29,295	anchors on the office's stated date where one exists; describes what customers receive, not rule correctness; corrected in v1.1 (section 5.2)
Schedule the API serves, production-weighted	99.99%		estimated over 9,166,622 production records from n=29,295	as above
Arm B event consistency, computed records	94.70%	[94.41, 94.98]	24,091 events	3,471 events on declared records not scored
Arm B event consistency, served schedule	92.71%	[92.39, 93.01]	27,562 events	no projected term past a lapsed office-stated expiry (section 5.3)
Arm C opposition consistency	90.99%	[90.62, 91.35]	23,751 verifiable oppositions	Wilson interval, not design-based (section 4.2); pinned per cell in CI

Metric	Value	95% CI	n	Notes
Disagreements remaining after the declared cohorts	95		0.35% of computed	all classified, section 5.5
Genuinely unexplained residual	3 records			
v0.91 full-population exact agreement (engine guesses scored against it)	96.08%	[96.00, 96.15]	282,393 (expanded same-seed sample)	retained for continuity; 96.00% on the committed sample
v0.91 modeled population (post hoc)	99.74%	[99.72, 99.76]	282,393	superseded by the declared policy, which specifies the same three cohorts in code

The primary version 1.0 result is the verified-or-declared figure: every comparable record is either computed and exactly equal to the office's published date, or explicitly declared as requiring that date. The engine correctness figure isolates the arithmetic: on the records the engine claims to compute, with the office's date withheld, it is 99.65% exact. The served figures describe what a docketing customer experiences and are partially self-referential: the served schedule reproduces a date the office published, so 99.99% there is close to circular by construction and is reported as a description, not as evidence. The v0.91 full-population figure is retained so the change of framing is visible: it scored the engine's guesses on the declared cohorts, which the engine no longer makes. The expanded 306,896- record stability check of v0.91 is not re-run here; its purpose (rate stability at depth) was met and the declared policy removes the cohorts that drove its difference from the committed sample.

5.2 Arm A by office and route (pure computation, office's date withheld, version 1.1)

Committed sample; exact agreement with the office-stated expiry on computed records, with the declared count per cell.

Office	Route	n computed	Exact agreement	Declared
EUIPO	direct	2,000	100.00%	0
EUIPO	madrid	2,000	99.90%	0
WIPO	madrid (IR and designations)	1,248	100.00%	0
Switzerland (IPI)	direct	2,000	99.90%	0

Office	Route	n computed	Exact agreement	Declared
Switzerland (IPI)	madrid	2,000	99.80%	0
Sweden (PRV)	direct	1,505	99.60%	0
Sweden (PRV)	madrid	1,500	100.00%	0
Norway (NIPO)	direct	2,000	99.35%	0
Norway (NIPO)	madrid	2,000	100.00%	0
Canada (CIPO)	direct	2,000	99.65%	0
Canada (CIPO)	madrid	500	100.00%	0
France (INPI)	direct	2,000	98.40%	0
USPTO	direct (derived expiry)	1,502	99.80%	0
Australia (IPAU)	direct	1,637	99.94%	363 (filed before 1996-01-01)
Australia (IPAU)	madrid	1,500	99.80%	0
Singapore (IPOS)	direct	1,408	98.44%	592 (filed before 1999-01-15)
Singapore (IPOS)	madrid	0	(declared)	1,540

Three cells moved materially against v0.91, all by mechanism: Australia direct (84.90% to 99.94%) and Singapore direct (82.75% to 98.44%) because the legacy-cadence cohorts are now declared instead of guessed; and the Madrid cells at Canada (90.60% to 100.00%), Australia (89.20% to 99.80%), Norway and Sweden because designations now anchor on the WIPO parent's international-registration date (209 former disagreements, all resolved exactly). Singapore's Madrid designations are declared in this arm: IPOS is not an office where the designation's own dates were verified as the IR date, and its priority-linked international marks carry no anchor on the row. On the served schedule (office expiry present) those 1,540 records and the 2,000 Singapore direct records all reproduce the office's date (100.00%).

Correction in v1.1 (served schedule). Versions up to 1.0 reported the served figures as 97.01% (Singapore Madrid) and 97.90% (Singapore direct), 99.65% overall, and attributed the Singapore residual to the register-error class of section 5.5. That was wrong: it was a

scoring error in our evaluation tool, not in the engine or the register. When the engine moves a renewal date off a weekend or office holiday, the evaluator recovers the un-moved anniversary before comparing, so a lawful roll is not counted as a disagreement. On the served path the schedule is anchored on the office's own expiry, so the next due date falls in the same year as that anchor, and the evaluator skipped the recovery step for exactly that case. A correctly rolled date was compared with the un-rolled office expiry and scored as a miss. 88 of the 103 served disagreements in version 1.0 were this error, all at Singapore. Fixed in the evaluator (with a test written to fail first); the served schedule now reproduces the office's date for 29,292 of 29,295 records, and the 3 that remain are USPTO derived-expiry rows. The pure computation, every engine-correctness figure, Arm B and Arm C were not affected: the pure run is identical with and without the fix. The recovery accepts any forward shift of up to 20 days (the engine's rollover bound), so in principle it could excuse a date that is late by a few days. We checked every row it recovers in the v1.1 served run, 4,540 including the 225 this correction added: in each one, every day from the anniversary to the served date is a closed day on that row's office calendar and the served date is open, so each is a lawful weekend or holiday roll (1 to 12 days).

The Sweden direct cell moved from 99.93% to 99.60% because the Swedish "fix" of section 5.4 was reversed (it is withdrawn there). WIPO now counts 1,248 records, 766 international registrations and 482 designations, as more designated jurisdictions became computable; all agree.

5.3 Arm B by office and route (events against windows, version 1.1)

Office	Route	Events (computed records)	Pure	Events (all)	As served
USPTO	direct	4,426	99.66%	4,426	99.66%
USPTO	madrid	1,956	99.74%	1,956	99.74%
Australia (IPAU)	direct	3,308	99.67%	6,779	90.50%
Switzerland (IPI)	direct	2,014	99.40%	2,014	99.40%
Australia (IPAU)	madrid	1,830	95.79%	1,830	89.13%
Sweden (PRV)	madrid	448	97.54%	448	97.54%
France (INPI)	direct	2,993	90.34%	2,993	90.78%
Sweden (PRV)	direct	5,006	87.91%	5,006	87.91%
Canada (CIPO)	direct	2,087	87.97%	2,087	88.36%

The USPTO cells are the cleanest test of the most intricate rules in the system (Section 8, 9, and 71 declaration windows including grace): 99.7% of 6,382 independent filing events fall inside the computed windows. The Australian direct cell, 57.62% in v0.91, is 99.67% once the 3,471 events on declared pre-1996 records are set aside (the engine claims no window for them); on the served schedule, which anchors those records on the office's expiry, the same events verify at 90.50%. The Australian Madrid cell fell from 98.25% to 95.79% when its renewal window was clamped to the six months the International Bureau allows (Madrid Common Regulations Rule 30(1)(b)) instead of IP Australia's twelve: 45 events now fall 6 to 12 months before the due date. This is not established as an accuracy loss. Outside USPTO the evaluation admits every renewal-type event, including reminders and office notifications, and Rule 30(1)(b) itself accepts early payment by deeming it received at the six-month point; the cell is reported as measured.

On the served schedule the same Australian Madrid cell is 89.13% (1,631 of 1,830 events). Since October 2026 the engine no longer projects a further renewal term past an office-stated expiry whose grace and restoration windows have passed with no renewal on record; it reports that the renewal is not on record instead of assuming one was made. IP Australia keeps the original term end on Madrid designations that were renewed through WIPO, so the served schedule also reads the expiry WIPO published for the international registration as evidence of that renewal. The evaluation supplies that evidence from the WIPO-parent join committed with the artifacts. 168 registered designations in the sample state an expiry between 2003 and 2016 while the register records a renewal about one cycle later. For the 47 of them whose WIPO registration shows a later expiry, the 61 renewal events stay inside a served window. The other 121 designations are not in that join, which was built in July 2026 for a different purpose (designations whose international-registration date was missing from the row), so the evaluation has no WIPO evidence for them either way: their 122 events fall after the last served window and score as misses. We report the cell as measured. One France direct event moves the same way (90.81% to 90.78%): a national mark whose record still states a 2015-07-16 expiry, although the register's events show renewals in 2015 and 2025; the engine reads the stated date only (it does not read event history). The pure computation takes no office expiry and is unaffected.

The remaining misses cluster by mechanism. Sweden direct: 470 of 605 misses are filings BEFORE the computed window opened (median 123 days early), consistent with the twelve-month renewal window of the pre-2019 Swedish Act, which is not modelled (the same class as Norway's pre-2024 window, section 7). France: 261 of 289 misses are recorded AFTER grace expiry (median 119 days), a register recording lag rather than a window error. Canada: 153 of 251 misses are dated about 9.6 years before the computed window (median 3,495 days), that is, on a ten-year cadence where the legacy ladder computes fifteen; this residual is filed. A variant metric that also counts an event dated exactly at a computed cycle anniversary changed no cell: the "renewed-until anniversary" recording convention hypothesized in v0.91 does not explain these misses, and that sentence is withdrawn.

5.4 Defects the verification found in our own rules, all fixed

Each fix was verified against primary legal sources and required to fit the register data with no regressions in any other cell. Pre-fix and post-fix agreement is reported for each.

Finding	Statutory basis	Effect of the fix
1. The WIPO configuration was keyed to a different code than production records carry, so every international registration silently received zero computed deadlines	Registry alias, no rule change	766 newly covered IR records agree at 100.00%
2. Canadian legacy renewal cadence: the 2019 reform shortens each term only when the period starts on or after the cutoff, so pre-2019 marks renew on successive 15-year terms first	Trademarks Act (Canada) s.46, in force 2019-06-17	CIPO direct Arm A 73.15% to 99.65%; Arm B 64.73% to 86.97%
3. Madrid designations renew on the international registration's ten-year anniversary, not the national trigger	Madrid Protocol Art. 6(1), 7(1)	CIPO madrid 0.00% to 90.60%; Norway madrid 45.75% to 99.90%; EUIPO madrid 70.30% to 99.90%
4. The German end-of-month rule leaked onto German Madrid designations	End-of-month rule applies to national registrations only	26 records corrected, up to 30 days
5. Withdrawn in v1.1. Earlier versions reported that the Swedish 2019 trigger reform was keyed on the filing date instead of the registration date, and "fixed" it	Varumärkeslag (2010:1877), transitional provisions SFS 2018:1652 p. 4 a: the former term rules continue to apply to applications made before 2019-01-01	Not a defect: the filing-date key was right in law and the change was reversed. The 5 records it had moved are disagreements again, where the Swedish register states a date the transitional provision does not support (section 5.5)
6. US Section 71 windows were anchored on the protection-statement date; the statute keys every Section 71 window to the certificate issuance date, the US registration date	15 U.S.C. 1141k(a); 37 CFR 7.36(b); TMEP 1613.04	USPTO madrid Arm B 66.16% to 99.59%

Finding 5 stays in the table, marked withdrawn, so its history remains visible: the transitional provisions keep the registration-anchored term for every application made before the reform, whenever it was registered, so the original filing-date key was correct. Findings 1 to 4 and 6

stand.

Finding 6 deserves a note on method: an anchor sweep over 1,956 observed Section 71 filings scored the registration date at 99.39% against 66.16% for the previously used field and at most 64% for every other candidate, and the statutory text, pulled independently, names the same date. Data and statute converged before the rule changed.

5.5 Classification of every remaining disagreement

All 544 in-scope structural disagreements at expanded scale were classified mechanically, each against production event and filing history, with every claimed engine defect adversarially re-verified:

Class	Count (vo.91, expanded sample)	Disposition in version 1.1
Classified by Signa as a register-data error, with per-record evidence	311	Still counted against the computed population; the 95 committed-sample disagreements that remain are this class (France renewals recorded as new deposits under the former law, 32; Singapore direct post-1999, 22; the Norwegian bulk-load placeholder expiry, 13; Canada, 7; Switzerland, 6; Sweden, 6; Australia, 4; USPTO derived, 3; EUIPO, 2)
Anchoring date absent from the record (Madrid subsequent designations, Singapore priority-linked international marks)	225	Resolved by mechanism: designations with a WIPO parent now anchor on its IR date (exact); IPOS designations without an anchor are declared; the served schedule anchors on the office's expiry
Records mis-routed by ingestion (Madrid designations ingested as direct)	5	Ingestion fix shipped; classification, not rule, error
Genuinely unexplained	3	Open

Zero claims survived adversarial verification as a true rule-logic defect: every cohort initially flagged as an engine bug reduced to a missing input or a register error. The register-error class includes French renewals recorded as new deposits under the former law, a Norwegian bulk-load placeholder expiry shared by unrelated marks, and ancient Canadian statutory-anchor phase offsets. Registers are large and contain errors; that a verification of this kind surfaces them, with evidence, is the strongest argument for computing deadlines rather than echoing stored ones.

5.6 Opposition windows against 23,751 observed oppositions

Office	Route	Verifiable n	Consistency
Canada (CIPO)	madrid	1,368	99.85%
EUIPO	direct	4,000	99.72%
EUIPO	madrid	4,000	99.05%
Australia (IPAU)	madrid	1,500	94.80%
Canada (CIPO)	direct	4,000	92.53%
Australia (IPAU)	direct	1,839	90.54%
Singapore (IPOS)	madrid	1,446	87.21%
Singapore (IPOS)	direct	3,991	76.05%
USPTO	direct	1,504	75.27%
USPTO	madrid	103	75.73%
Pooled		23,751	90.99%

Five cells moved since version 1.0. The EUIPO cells rose because closes the engine had served un-rolled are now rolled: weekends roll even beyond a calendar's pinned coverage, and EUIPO's 2023 and 2024 closure decisions are now pinned evidence. The two Singapore cells moved by a few rows for the same weekend-roll reason. The Canada direct cell moved from 90.83% to 92.53% because of a change to the evaluation tool, not to the rule: the tool's benchmark for Canada's routine extension now uses the extension length in force when each window closed, which moved 68 oppositions from beyond the window to inside the extension.

This arm found two more defects of the silent-mismatch kind, both fixed. The first is a rule defect (the one opposition-rule defect, which brings the rule-defect total to six: the five maintenance-rule defects of section 5.4 plus this one); the second is an ingestion defect. (7) The IP Australia opposition rule was keyed to an office code production records do not carry, so every Australian record served a null opposition window until this evaluation noticed. (8) The Singapore connector stored the latest gazette republication as the record's first publication date (the office republishes after amendments and registration), which made 749 sampled oppositions appear to have been filed before publication. Every one of them falls inside the two-month window of the earliest gazette entry; the ingestion fix applies to newly synced records, with a backfill of existing records filed. The evaluation also drove three rule additions verified against it: the Australian window's era split (three months before the 2013-04-15 reform, two after, with the routinely recorded one-month structural extension for the two-step filing process), and dedicated Madrid-designation opposition rules at CIPO, IP

Australia, and IPOS, of which only Canada's differs from the domestic rule (a single extension hard-capped at four months by Trademarks Regulations s.125). The Canadian Madrid cell then verifies at 99.85%, with 2 of 1,368 oppositions beyond the statutory cap.

The USPTO cells are depressed by a known data property, not the window arithmetic: proceedings data attaches an opposition to both parties' records, the registration-timing filter can only remove the opposer-side rows it can prove, and the residual splits roughly half extension-boundary cases at the 180-day statutory ceiling and half remaining wrong-side linkage (real, decided oppositions attached to a mark whose single publication was years earlier, which no opposition window can contain). The linkage correction is filed as ingestion work; the Singapore direct cell will be re-measured after the publication-date backfill.

6. Analysis

Misses cluster by mechanism, not randomly. Every material disagreement cohort in this evaluation bucketed cleanly by office, route, or era: a legacy cadence, a wrong anchor column, a keying mismatch, a data-linkage property. Nothing material diffused randomly, which is what one expects if the arithmetic is right and the residual causes are structural. The off-by-N distribution says the same thing: in the v1.1 run, 24 of the 26,800 computed-record comparisons (0.09%) are off by one to seven days (17 by one day, 7 by two to seven); the other 71 disagreements land more than a week off, most of them months or years.

The office's date and the statute check each other. The pure computation and the office-stated expiry agree on 99.65% of the records the engine computes (v1.1), which means each is a working audit of the other. Where they disagree, adjudication attributed the disagreement to the register more often than to the engine (311 records classified as register-data errors, against no further rule defect identified among the investigated residuals). This is the argument for computing deadlines from statute rather than echoing the register: the computation catches register errors, and the register catches rule errors, and this report is the record of both directions working.

Recomputation at read time beats stored deadlines. Six rule defects were fixed during this evaluation (five in the maintenance rules, one in the opposition rules). Because deadlines are computed at read time from a pure function, every fix applied to every record instantly, with no re-audit of stored calendars. The before-and-after numbers in section 5.4 were produced by re-running the same frozen sample through the corrected engine, which is exactly what production did the moment each fix deployed.

Arm B is asymmetric, and we score it conservatively. An observed filing inside our window proves the window was usable in practice; the absence of an event proves nothing, and an event recorded outside the window is scored against us whatever its cause. The residual misses have mechanisms (section 5.3): early filings under a pre-reform Swedish window we do not model, late-recorded French renewals, and a Canadian legacy-cadence

residual. None is excluded. Version 0.91 attributed part of the residual to a "renewed-until anniversary" recording convention; a metric variant that credits exactly that convention changed no cell, so that explanation is withdrawn.

Declaring is a feature. The largest change in version 1.0 is not a number but a behavior: the engine now states, per record and with a reason, when it needs the office's date instead of computing one. For the three cohorts concerned, the register had shown the computed guess to be wrong often enough that serving it was a liability; a docketing system that says "I need the office's date here" is more useful than one that is confidently wrong one time in six. Because the same rule is specified in the product and applied in the evaluation, the population the engine is scored on is, for the first time, exactly the population it claims to compute.

7. Limitations

- **Coverage is not universal.** For offices whose deadline rules are not yet modeled, the API says `supported: false` rather than guessing, and this report claims nothing about them. This report verifies the ten offices of section 4.2 only.
- **8.5% of comparable records are declared, not computed.** The engine states that it needs the office's date for Australian direct marks filed before 1996, Singapore direct marks filed before 1999, and Singapore Madrid designations. In production every one of those offices states an expiry, so the served schedule anchors on it; the declared share is a limit on the pure statutory computation, disclosed rather than hidden. Extending Singapore's Madrid coverage requires the WIPO-parent join to reach its priority-linked international marks.
- **Madrid stand-ins are verified per office, not universal.** Where no WIPO parent is found, a designation's own date is used as the IR date only at the seven offices where this evaluation verified that equivalence; elsewhere the engine declares.
- **The USPTO Arm A cell is derived, not office-stated.** USPTO publishes no expiry date, so that one cell shares derivation logic with the thing it tests. The USPTO evidence that matters is Arm B: 99.7% of 6,382 independent filing events inside computed windows.
- **WIPO renewal evidence is partial in the served Australian Madrid cell.** The served schedule reads the expiry WIPO published for a designation's international registration as evidence that IP Australia's lapsed stated expiry was renewed. The evaluation supplies it from a WIPO join built in July 2026 for another purpose, which covers 47 of the 168 designations concerned; the other 121 have no evidence either way, so their 122 events count against the 89.13% served cell (section 5.3). That join also carries no WIPO status for the designated territory, which the product uses to withhold the evidence when the territory's designation ended.

- **Four offices have no event ground truth.** EUIPO, IPOS, Norway, and WIPO emit no renewal events in our pipeline, so their results rest on Arm A only.
- **Opposition verification is partial where the data is.** The USPTO opposition cells are bounded by a proceedings-to-record linkage property described in section 5.6, and the Singapore direct cell awaits the publication-date backfill. The EUIPO cells (n=8,000, no extension regime to absorb error) are the cleanest external check of the opposition engine.
- **Maintenance-deadline holiday rollover is opted in by 3 configurations.** Opposition windows apply office holiday calendars across all covered offices; on the maintenance side ten calendars are mapped but only the USPTO (domestic and Madrid) and IPOS configurations opt into rollover, and it is a small number of days at window edges elsewhere. (v0.92 correction: v0.91 said "wired for 10 offices".)
- **Some calendar data is forward-looking.** Rule-computed office holiday calendars run to 2100 (v0.92; previously 2050); the lunar, Islamic and administratively-set calendars (EUIPO, IPOS, IPOPHL, DIP, IP Vietnam, JPO, KIPO, CNIPA) are curated through 2030; a small number of future entries are projected pending official publication and marked as such in the data.
- **One office's renewal trigger is approximated.** The Philippine renewal trigger is computed from registration cycles where the underlying renewal event is not itself carried on the record.
- **The list-endpoint deadline filter is an approximation.** Filtering trademark lists by upcoming deadline uses a stored renewal-due column; the authoritative computation is the detail endpoint and the compute endpoints, which is what this report verifies.
- **Ground truth is the register, and registers contain errors.** That cuts both ways: Arm A disagreements are individually adjudicated rather than assumed to be our fault, and the register-error class in section 5.5 is evidence-backed per record, not a convenient bucket.
- **Rule verification is dated, not continuous, but regression is gated.** Every rule carries the date it was last verified against sources (the configurations touched by this evaluation carry 2026-07-11 or later). Statutes change; the report version will be bumped when rules or measurements are refreshed. Between refreshes, the frozen-sample regression gate of section 3.5 prevents silent movement.

8. Reproducibility and versioning

- The sample manifests (seeded queries and SHA-256 record-identifier digests) were frozen before comparison ran. The evaluation harness, metrics, frozen samples, per-run results, and the mechanism-level classification of every disagreement are committed artifacts, and the headline counts are pinned in continuous integration against the frozen sample.

- The rule corpus this report verifies is publicly inspectable, with statutory citations, via `GET /v1/deadline-rules` and `GET /v1/opposition-rules`.
- The artifacts are published at <https://github.com/signa-so/research> (folder `deadline-verification`): the frozen manifests, the samples, the events, the oppositions, and the per-record comparison results, so the adjudication of every disagreement can be checked, not taken on faith.
- Source data is public register data; no customer data appears in any evaluation set.
- Published numbers are never edited in place; updated measurements ship as a new version with a changelog entry.

Changelog

- **v1.1 (2026-10-03, served event figures re-measured 2026-10-04):** Measurement refresh and two corrections; no change of method or policy. Every figure re-run on the same frozen samples at the same as-of date (2026-07-07) against the current rules. Moves: verified or declared 99.69% of 29,102 to 99.68% of 29,295 (95% CI 99.60 to 99.73); computed records 26,607 (91.4%) to 26,800 (91.5%); pure engine correctness 99.66% to 99.65% (95% CI 99.57 to 99.71), production-weighted 99.82% unchanged; declared 2,495 unchanged (8.57% to 8.52% of comparable); Arm B on computed records 94.71% to 94.70% (USPTO 99.4% to 99.7%; Australia Madrid 98.25% to 95.79% after the Madrid-route renewal window was clamped to Rule 30(1)(b)'s six months, not established as an accuracy loss; France 90.11% to 90.34%; Canada 86.97% to 87.97%); Arm B served 93.18% to 92.71%; Arm C pooled 90.23% to 90.99% (EUIPO 98.22% to 99.72% direct and 97.95% to 99.05% Madrid; Singapore 75.90% to 76.05% direct and 87.00% to 87.21% Madrid; Canada direct 90.83% to 92.53%, an evaluation-tool change to the era of the extension benchmark, not a rule change); disagreements after the declared cohorts 90 to 95; WIPO Arm A rows 1,049 to 1,248, all agreeing; Sweden direct Arm A 99.93% to 99.60%. Section 5.3 miss clusters re-derived: France 261 of 289 after grace (was 268 of 296), Canada 153 of 251 about 9.6 years early (was 171 of 272). Correction 1, served schedule: an evaluation-tool scoring error compared correctly rolled due dates with unrolled office expiries on the served path; fixed, the served schedule now reproduces the office's date for 29,292 of 29,295 records (99.99%, production-weighted 99.99%), against 99.65% and 99.89% reported before, and the 97.01% and 97.90% Singapore served figures, previously attributed to register errors, are withdrawn (section 5.2). Correction 2, Sweden: the "Swedish 2019 trigger" rule defect is withdrawn; the original filing-date key was correct under the transitional provisions of SFS 2018:1652 (p. 4 a), the change was reversed, and the rule-defect total is six (five maintenance, one opposition); sections 1, 3.2, 5.4, 5.5 and 6 corrected accordingly. Section 7: the named examples of unmodeled offices were removed; Japan, Korea and China are now modeled. Sections 1 and 3: jurisdiction, rule and calendar counts are dated to the v1.0 evaluation, with the October 2026 counts beside them. Section 6: the off-by-N and agreement figures are the v1.1 run's

(0.09% of comparisons off by one to seven days and 99.65% agreement, where earlier versions quoted 0.38% and 99.74% from the July 2026 runs). Section 5.2: every row the fixed scorer recovers was checked to be a lawful roll. Re-measured before external publication (2026-10-04): the v1.1 figures drafted on 2026-10-03 predate the October 2026 engine change that stops projecting a renewal term past a lapsed office-stated expiry, and the WIPO renewal evidence the served schedule reads with it. The evaluation now supplies that evidence from the committed WIPO join. On the released engine served Arm B moved from 93.15% (25,675 of 27,562) to 92.71% (25,552; 95% CI 92.39 to 93.01): Australia Madrid served 95.79% to 89.13% (1,753 to 1,631 of 1,830) and France direct served 90.81% to 90.78% (2,718 to 2,717 of 2,993), explained in section 5.3 and limited as section 7 states. No other figure moved: the pure computation, the served Arm A and Arm C are identical. The documentation page, published on 2026-10-04 with 93.2%, was corrected to 92.7%.

- **v1.0 (2026-08-30):** Declared computability. The engine declines, with a stated reason (`requires_office_date`), where a record lacks the statutory input and no office expiry anchors it (AU direct filed before 1996-01-01, SG direct filed before 1999-01-15, Madrid designations with no international-registration anchor); Madrid designations now anchor on the WIPO parent registration's date where one is ingested (209 former disagreements resolved exactly), and designation-date stand-ins are used only at the seven offices where they were verified. The post hoc "modeled population" is replaced by this pre-specified policy, applied identically in the product and the harness. Headline re-framed: verified or declared 99.69% of 29,102; pure engine correctness 99.66% on computed records (99.82% production-weighted); served schedule 99.65%/99.89% (unchanged); Arm B 94.71% on computed records (93.18% served); 90 disagreements remain, all classified; 3 unexplained. Arm B tables and the section 6 Arm B analysis rewritten (the "renewed-until anniversary" explanation is withdrawn after a direct test). Rule-computed holiday calendars extended to 2100. The v0.91 full-population figure is retained in section 5.1 for continuity.
- **v0.92 (2026-08-30):** Hardening pass, no re-measurement of the headline. A mutation-testing hardening run over the rules package (57 files, 2,946 mutants, 100% killed or classified) and the compute endpoint produced 34 findings and a fix wave. None changes an Arm A or Arm B date on the ten verified offices; one deliberate pin move: the Arm A pure count moved from 27,931/29,096 to 27,937/29,102 because six WIPO-sample rows with jurisdiction MX now compute. Those six carry a filing date before the 2020-11-05 cutoff and no grant date, so once the Mexican 2020 era split is keyed on the filing date (LFPPI Transitorio DÉCIMO, verified against the live IMPI register) they resolve to the pre-LFPPI filing anchor instead of having no resolvable trigger at all. All six agree with the office-stated expiry. Arm C is now evaluated at the frozen as-of date and pinned per cell in CI; the pin reproduces the v0.91 §5.6 cells exactly (pooled 90.23%). Rule corrections outside the verified offices: Danish 2019 era split keyed on the registration date (Lov 1533/2018 §8 stk. 3); Philippine third-year DAU window runs from the filing date (RA 8293 §124.2); Thai opposition 90-day era before 2016-07-28. Engine semantics:

the due day and the last grace day now read as actionable (matching the opposition engine); expired/cancelled marks keep their open late-renewal and restoration windows; restoration is derived from missed renewals; the opposition coverage-skip threshold shares the rollover bound (20 days, was 14). §7 corrected: maintenance rollover is opted in by 3 configurations, not "wired for 10 offices".

- **vo.91 (2026-07-12):** Framing corrections after an external methodological review. Full-population agreement (96.08% expanded scale, 96.00% committed small sample) is now the primary headline; the 99.74% figure is labeled a post hoc modeled-population result with its exclusions named, and the 99.89% figure is labeled office-date-anchored and partially self-referential. Removed the "holdout" and "confirmatory" framing (the expanded run is a same-seed superset stability check), narrowed the CI regression-gate claim to what it actually pins, disclosed the 15-to-45-year horizon protocol amendment, reconciled the rule-defect count at seven (six maintenance plus one opposition), softened first-party classification language, added a first-party-evaluation disclosure, and wired the published artifact links.
- **vo.9 (2026-07-12):** All results populated from the completed evaluation: pre-registered 31,547-record sample, 306,896-record expanded same-seed sample, 34,132-opposition sample. Six maintenance-rule fixes, one opposition-rule fix, two ingestion fixes, and three rule additions documented with pre-fix and post-fix numbers. Engine sections re-verified against the shipped codebase (23 maintenance configurations, 27 opposition rules, 24 holiday calendars, approximately 346 rules-package tests).
- **vo.1-draft (2026-07-07):** Internal draft. Engine sections verified against the codebase; results pending the frozen evaluation run.

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